

5 Inorganic Nanocarriers

Gold Nanoparticles, Iron Nanoparticles, Silica Nanoparticles, Quantum Dots

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5.1 INTRODUCTION

Inorganic nanocarriers have gained significant importance in biomedical applications because of their unique, tailorable characteristics, which are impossible in organic forms. These features include better stability, facile functionalization, and distinctive optical, magnetic, and electronic properties. Some of the significant nanocarriers are gold nanoparticles (AuNPs), iron nanoparticles (FeNPs), silica nanoparticles (SiNPs), and quantum dots (QDs).

AuNPs have outstanding optical properties due to surface plasmon resonance (SPR), allowing them to absorb and scatter light. Such a property is essential for imaging and diagnostics. Moreover, attaching different ligands and biomolecules to their surfaces has increased biocompatibility and augmented targeting specificity in their application in drug delivery systems. Initial studies done by El-Sayed et al. in 2005 illustrated that through light, the AuNPs could be converted to heat with their selective propensity to destroy cancer cells, making them multifunctional for simultaneous imaging and therapy (1).

Superparamagnetic FeNPs, especially superparamagnetic iron oxide nanoparticles (SPIONs), have revolutionized magnetic resonance imaging (MRI) due to their better magnetic behavior. SPIONs offer significant enhancement in contrast to MRI, thus improving the diagnosis of diseases. Their magnetism is applied in magnetic hyperthermia to cause localized heating for cancer treatment. This application was introduced by Weissleder et al. in 1990. Studies are underway and are still investigating the potential of the same SPIONs used as vectors for targeted drug delivery and theranostics (2).

The SiNPs are reputed to be biocompatible. They offer an extensive surface area. In addition, they provide a simple approach to their surface functionalization. Such properties make them excellent for use in the delivery of drugs as the pores can be loaded with therapeutic agents and designed for a specific target. Furthermore, high drug payload ability in mesoporous silica nanoparticles (MSNs) was shown to enhance therapeutic efficacy by Slowing et al. in 2008, making it one of the forces compelling a broad interest in research applications in cancer therapy and other medical fields (3).

QDs are very promising for bioimaging and diagnostics, owing to their size-tunable fluorescence and high photostability. These semiconductor nanoparticles yield very bright and long-lasting signals, which are critical for prolonged imaging experiments. The groundbreaking work in 1998 by Chan and Nie opened the way for broad applications of QDs in monitoring cellular processes and identifying disease markers. Studies have been conducted with the hope of reducing QD toxicity and establishing biocompatible coatings in pursuit of safer medical applications (4).

Generally, the distinctive features, from the optical features of AuNPs and the magnetic capabilities of FeNPs to their large surface area and biocompatibility of SiNPs, along with QDs' fluorescence, make them outstanding candidate materials for many biomedical applications. These functions render advanced imaging, precise drug delivery, and a novel approach toward therapeutics, with the possibility of moving the field of nanomedicine further ahead.

In the field of drug delivery, inorganic nanocarriers such as AuNPs, FeNPs, SiNPs, and QDs offer several advantages over their organic counterparts, including polymers and liposomes. These advantages are primarily a result of their inherent physical and chemical properties, which make them efficient in delivery and, most importantly, stable. Furthermore, inorganic nanocarriers have unique surface properties and high surface-to-volume ratios that contribute to their delivery efficiency. For example, the AuNPs can be modified highly precisely at the surface to attach specific targeting ligands and drugs, thus enhancing their delivery precision to particular cells or tissues. Research such as that carried out by Jain et al. in 2012 has already shown that once conjugated with antibodies, AuNPs can target and effectively deliver drugs specifically into cancer cells (5). Among the organic nanocarriers, premature drug release and low targeting selectivity compared to their inorganic counterparts prevent high delivery efficiency.

Another critical advantage is the stability of inorganic nanocarriers under physiological conditions. Materials such as iron oxide and silica maintain the structure of the nanocarriers and are resistant to enzymatic degradation. FeNPs, in particular, can retain their magnetic properties and structural stability, which would be required for their use in MRI and target drug delivery. Biodegradation of organic carriers, such as liposomes, due to enzymes or instability in the blood can reduce the therapeutic effect.

In the case of individual types of inorganic nanocarriers, their functional integrity is increased by enhancing chemical stability and resistance to oxidation by AuNPs during circulation and at the target site.

Core-shell structures and surface coatings are recent innovations that further stabilize AuNPs and add new functionalities (6). The magnetic properties and stability of FeNPs, especially SPIONs, make them a widely popular product. Coatings with dextran or polyethylene glycol (PEG) make FeNPs more biocompatible and help them not to aggregate, a stabilization strategy that is less frequently needed for organic carriers because they are not intrinsically stable against aggregation, unlike inorganic materials (7).

Thus, SiNPs are very stable and possess a high surface area, which makes them very appropriate for making drug delivery systems. This is explained by the porous structure of mesoporous SiNPs, which ensures their effectiveness in high drug loading and the controlled release of substantial drug amounts. In this regard, several studies indicate that MSNs are stable and functional across various biological environments and, in general, outperform many other organic carriers for sustained release and protection against premature degradation (8).

Lastly, QDs are helpful for their unique optical properties and stability, making them applicable to imaging applications. As they are inorganic by constitution, they resist photobleaching and chemical degradation, which otherwise ensures reliable performance in studies where imaging is carried out over a long period of time. Organic fluorescent dyes, on the contrary, are relatively resistant to these situations and render poor performance in terms of prolonged imaging scenarios (9).

Inorganic nanocarriers are much superior to other existing drug delivery systems because of their outstanding physicochemical properties. Problems of stability, loading capacity, and targeted delivery are generally encountered in traditional systems like liposomes and polymer-based carriers. Inorganic nanocarriers, such as AuNPs, IONPs, SiNPs, and QDs, have inherently stable structures, versatile surface chemistry, and multifunctional abilities.

One of the most significant advantages of inorganic nanocarriers is enhanced stability in biological environments. Traditional carriers are easily degradable, making them weak and prone to leakage and degradation, thus causing premature drug release. The inorganic nanoparticles are more stable, do not get damaged easily under physiological conditions, such as enzymatic degradation, and do not disrupt their structure. For instance, SiNPs are well known for their robust inert matrix, which sustains the delivery of drugs without causing premature drug release (10).

In this way, inorganic nanocarriers offer controlled drug release kinetics and targeted delivery. By functionalizing the AuNPs with different types of targeting ligands, they can specifically target specific cells or tissues through either ligand-receptor interactions or physical entrapment. This would minimize off-target impacts and enhance therapeutic effectiveness. By way of an example, Dykman and Khlebtsov in 2012 showed that AuNPs could deliver chemotherapeutics directly into cancer cells so that systemic toxicity is reduced (11).

Moreover, inorganic nanocarriers are inherently multifunctional and allow simultaneous imaging, therapy, and diagnostics. For example, QDs present particular optical properties that have been applied in fluorescence imaging, and they can be easily modified to carry therapeutic agents. These very features make tracking drug delivery in real-time possible, along with monitoring therapeutic outcomes (12).

Another crucial benefit, therefore, is the high drug-loading capacity of inorganic nanoparticles due to their large surface area-to-volume ratio. Large pore volumes of mesoporous SiNPs allow significant amounts of drugs to be encapsulated, providing high therapeutic payloads compared to traditional carriers. This will be essential in effective cancer therapy, for which potent doses of medicines are necessary (3).

Furthermore, inorganic nanocarriers enhance the delivery and stability of nucleic acids and proteins. These are critical but challenging areas in biomedicine due to the very nature of the instability of the biomolecules themselves and their susceptibility to degradation by nucleases and proteases. For example, SiNPs create a protective microenvironment that entraps the nucleic acid inside the porous structure and protects against nucleases, ensuring that the nucleic acid in its original form reaches the target site. Proteins can be similarly immobilized on or within inorganic nanocarriers to maintain their intact structure and functionality, thereby eluding the effects of denaturation and aggregation (13).

Cellular uptake for therapeutic molecules is further enhanced using inorganic nanocarriers. AuNPs can be functionalized by cell-penetrating peptides, which improve cellular uptake through endocytosis and provide therapeutic molecules access to intracellular compartments. Due to their small size and tunable surface properties, such nanoparticles can cross biological barriers and reach the target cells much more effectively than classical carriers (14).

Another important property is controlled release; for instance, MSNs can be designed to unload the drug in response to certain stimuli, like pH changes, redox conditions, or enzyme presence. This allows one to tune the kinetics of the release to the site of action to leave the least possible room for systemic side effects while at the same time achieving the highest possible therapeutic efficacy (3).

Last but not least, targeted delivery—be it gene therapy or protein-based therapy—the relevant delivery of the target to the affected cells will define the fate of failure and success of these interventions. For example, IONPs have been used to deliver siRNA to cancer cells with significant gene-silencing effects (15).

Therefore, such inorganic nanocarriers provide critical advancement over traditional systems of drug delivery that serve to increase therapeutic success and decrease side effects owing to their properties of stability, controlled and targeted delivery, multifunctionality, and high drug-loading capacity. In doing so, they will be essential in devising new and more effective therapeutic strategies for many diseases.

5.2 FUNDAMENTALS OF INORGANIC NANOCARRIERS

5.2.1 Basic Concepts and Properties

Several fundamental material properties are considered to have a profound influence on the effectiveness of inorganic nanocarriers in their biomedical applications. These directly depend on how nanocarriers act in biological environments, establishing stability, biocompatibility, drug-loading capacity, and the potential to effectively target specific cells or tissues.

The size and shape of inorganic nanoparticles play critical roles in biodistribution, cellular uptake, and clearance from the organism. Generally, nanoparticles range from 1 to 100 nanometers in size. Smaller nanoparticles are better at tissue penetration and less likely to be recognized and cleared by the immune system. Spherical nanoparticles are generally internalized by cells more readily than particles that are rod-shaped or irregularly shaped. Huang et al. in 2010 demonstrated that spherical AuNPs have a higher cellular uptake than their rod-shaped counterparts (16).

At the heart of interaction with biological systems lies surface chemistry. Coating modifications of the surface to be biocompatible, like PEG, helps reduce opsonization and recognition by the mononuclear phagocyte system (MPS) to increase the circulation time of the nanoparticle. Moreover, the functionalization of nanoparticles with targeting ligands (antibodies or peptides) can make drug delivery more selective and enable binding to a specific cell type. For instance, in 2011, Choi and co-authors showed that PEGylated IONPs significantly increased circulation time *in vivo* (17).

The core properties necessary for both imaging and therapy are magnetic and optical. Among the SPIONs, there are strong magnetic properties, and they are applied as a contrast agent in MRI. AuNPs have unique optical properties, such as SPR, that make them suitable for photothermal therapy and imaging. Similarly, QDs are preferable because of their size-tunable fluorescence, thus making them one of the most useful in bioimaging. These characteristics enhance the multifunctionality of the inorganic nanocarriers, therefore making them applicable in theranostics (18).

Notably, drug-loading capacity is an essential characteristic of drug release kinetics. Mesoporous SiNPs offer a large surface area and pore volume that can contain a high drug load. The release kinetics may be tuned with the possibility of changing surface chemistry or using stimuli-responsive materials that may provide a controlled release to minimize systemic side effects and maximize therapeutic effectiveness. Vallet-Regí in 2007 reviewed the potential of MSNs for controlled drug release (19).

Biocompatibility and toxicity are mainstays for *in vivo* applications. While both gold and silica are considered biocompatible, the safety of both components has been evaluated and documented to a high level. However, the long-term implications and toxicity, particularly of heavy metal-containing nanoparticles like cadmium found in QDs, must be studied. Designing safety and surface modification strategies is critical to minimizing potential adverse effects (20).

The common essential characteristics that make an inorganic material suitable for use as a nanocarrier bring about the delivery of therapeutic agents in a biocompatible manner and remain stable in biological environments. Structural stability is one crucial requirement; inorganic materials are, per se, reputed to maintain their structural integrity under physiological conditions, hence preventing premature degradation and fulfilling the objectives of successful delivery to the target site. For example, SiNPs possess excellent chemical and physical stress resistance properties applicable for sustained drug release (10).

The surface functionalization ability allows modification to attach different chemical groups, ligands, or biomolecules for the specificity of nanocarriers, thus further enhancing the efficiency of drug delivery. AuNPs can be functionalized with various thiol-containing molecules to significantly customize their surface properties (5). Controlled release mechanisms are typically accomplished using inorganic materials, such as mesoporous SiNPs. These nanoparticles can be altered to adjust the size and surface chemistry of their pores, enabling precise control over the rate at which drugs are released. Stimuli-responsive release mechanisms—pH- and temperature-responsive—are additional significant enhancements to develop novel targeted drug delivery.

Basic biocompatibility is a requirement. Materials such as gold and silica are well validated for their compatibility with biological systems, which makes it extremely important to establish safety in drug delivery

applications. Minimal toxicity and excellent compatibility with biological systems have been demonstrated for AuNPs (16) (Figure 5.1).

Multifunctionality enhances strategies for treatment. Inorganic nanoparticles can be designed to be multifunctional in terms of imaging, therapy, and diagnostics, all within the same platform. This is demonstrated in QDs, which have unique optical properties that offer the possibility of imaging combined with the delivery of a therapeutic agent. This added advantage is not available with organic carriers. All of these allow for more extensive options in diagnostics and therapy, such as MRI, photothermal therapy, and fluorescence imaging, among others.

5.2.2 Mechanisms of Action

Various advanced interaction mechanisms between inorganic nanocarriers and cellular environments bring out the effective delivery of therapeutic cargo. This occurs mainly due to the physical and chemical properties of the nanoparticles, such as size, shape, surface charge, and functionalization. In general, this delivery process consists of cellular uptake, intracellular trafficking, and finally, cargo release.

5.2.2.1 Uptake

Interactions between the inorganic nanocarriers and cells mainly start with their uptake through several endocytic pathways. The size and surface properties of the nanoparticles control the specific pathway employed. Smaller particles, 10–50 nm, are usually internalized more effectively through clathrin-mediated endocytosis, while larger ones >200 nm may be taken up in the cells by phagocytosis. Research by Chithrani et al. in 2006 reflected that AuNPs of different sizes follow varied endocytic pathways for entering the cell (21).

5.2.2.2 Intracellular Trafficking

Once the inorganic nanocarriers have entered the cell, they pass through the endosomes and lysosomes. If nanoparticles are prepared to show a response to pH changes, then acidic pH in these organelles would lead to the release of the cargo. For instance, pH-responsive MSNs may dissolve under acidic conditions, thus resulting in drug release in cancer cells, as demonstrated by Jiang et al. in 2013 (22).

5.2.2.3 Escape from Endosomes

To prevent their degradation inside the lysosomes, many inorganic nanocarriers are designed to escape from the endosomes and release their payload into the cytoplasm. This can occur through the proton sponge effect, where nanoparticles induce osmotic swelling and rupture of the endosome. For instance, cationic AuNPs can increase the concentration of protons in endosomes, thereby disrupting these vesicles and releasing their content into the cytoplasm (23).

5.2.2.4 Targeting Specific Organelles

Some inorganic nanocarriers are specifically designed to deliver their cargo to particular organelles like the nucleus or mitochondria. Functionalizing nanoparticles with nuclear localization signals (NLS) or mitochondrial targeting sequences (MTS) enables their targeted delivery to these critical organelles. For example, AuNPs conjugated with NLS peptides have demonstrated effective delivery of therapeutic agents directly into the nucleus (24).

The inorganic nanocarriers come close to cellular environments during endocytosis, intracellular trafficking, endosomal escape, and targeted delivery to specific organelles. These mechanisms could be assembled to ensure that the therapeutic cargo is effectively taken to suitable intracellular locations for maximum treatment.

Other studies about inorganic nanocarriers include the mechanisms by which targeted and controlled release of drugs is done to increase the precision and efficacy of drug delivery while cutting back on the side effects to yield more therapeutic outcomes.

5.2.2.5 Passive Targeting

It works through the enhanced permeability and retention (EPR) effect, where nanoparticles accumulate preferentially at the tumor sites due to leaky vasculature for cancerous growth. This increases the drug concentration at the tumor site, thereby increasing therapeutic efficacy, as postulated by Maeda et al. in the year 2000 (25).

5.2.2.6 Active Targeting

In the functionalization of nanocarrier surfaces, specific ligands such as antibodies, peptides, or small molecules with high binding affinities for receptors typically overexpressed on target cells can be used. This

Gold nanoparticles (AuNPs)



Physicochemical Properties
Easily functionalized
Suitable for gene and protein delivery

Biocompatibility
High biocompatibility
Inert nature
Low toxicity

Iron oxide nanoparticles (IONPs)



Physicochemical Properties
Useful for magnetic targeting
Imaging
Therapeutic delivery

Biocompatibility
Moderate biocompatibility
Potential for oxidative stress and inflammation

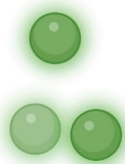
Silica nanoparticles (SiNPs)



Physicochemical Properties
Excellent controlled release properties

Biocompatibility
Biocompatible
Potential for pro-inflammatory and pro-oxidative effects

Quantum dots (QDs)



Physicochemical Properties
Mainly used in imaging and sensing due to optical properties

Biocompatibility
Significant biocompatibility challenges
Potential toxicity due to heavy metals

Figure 5.1 Key properties of inorganic nanocarriers. Gold nanoparticles (AuNPs): Surface plasmon resonance illustrated by oscillating electron cloud and electric field. Iron oxide nanoparticles (IONPs): Magnetic properties demonstrated by alignment under external magnetic field. Silica nanoparticles (SiNPs): Varying pore sizes affect loading capacity, crucial for drug delivery applications. Quantum dots (QDs): Size-dependent optical properties, with smaller particles emitting higher-energy (bluer) light.

allows drug delivery to have greater specificity for diseased cells, for example, in the targeted delivery of AuNPs conjugated with folic acid to cancer cells (26).

5.2.2.7 Magnetic Targeting

It makes use of magnetic nanoparticles, like iron oxide, which can be directed to a place in the body by an outside magnetic field; this method is most useful in tumor targeting (7).

5.2.2.8 pH-Responsive Release

Drugs bearing nanocarriers are engineered to release in response to the acidic levels of either tumors or intracellular compartments. MSNs with pH-responsive coatings are the best example. They dissolve in the acid surrounding the environment, hence releasing the drug (27).

5.2.2.9 Temperature-Responsive Release

This uses temperature-sensitive nanoparticles, which release drugs due to variations in local temperatures and are commonly used along with hyperthermia treatments. For example, AuNPs convert absorbed light into heat, thus releasing the drug in its vicinity (28).

5.2.2.10 Redox-Responsive Release

Li et al. in their study in 2020 demonstrated that high levels of glutathione cleave the disulfide linkages in the nanoparticles and release the drug; they also demonstrated the exact mechanisms used in enzyme-responsive releases (29). For instance, matrix metalloproteinase-sensitive nanoparticles release drugs in the presence of MMPs, a method that Yao et al. (2018) investigated in targeted cancer therapy (30).

As such, select targeting and well-thought-out overall mechanisms, followed by controlled release mechanisms through inorganic nanocarriers, assure effective and safe delivery of drugs toward overall medical treatment.

5.3 GOLD NANOPARTICLES

5.3.1 Synthesis and Properties

AuNPs are produced through various methods, which significantly impact their characteristics and render them valuable for a wide range of biomedical uses. Commonly used methods include chemical reduction, seed-mediated growth, green synthesis, and physical processes such as laser ablation.

5.3.1.1 Chemical Reduction

Out of all the techniques used, the most common one has been reducing gold salts, such as HAuCl₄, with agents like sodium citrate, sodium borohydride, or ascorbic acid. By adjusting the reducing agent's concentration, the temperature, and the reaction time during synthesis, AuNPs can develop customized sizes and shapes. In this process, the particles are usually produced spherically, although other shapes, such as rods or cubes, can be realized based on the nature of the reaction parameters. Turkevich et al. first described the essential background for this method of chemical reduction in 1951 with the citrate reduction method (31).

5.3.1.2 Seed-Mediated Growth

This process involves the formation of small seed particles that are synthesized from gold and then act as nucleation sites for further growth. It is a method in which control over nanoparticles' size and shape is readily possible. Surfactants, like cetyltrimethylammonium bromide (CTAB), enable the production of AuNPs in rod shapes with adjustable aspect ratios by adjusting the concentrations of the seed and growth solutions. Murphy et al. (2008) showed the versatility of this method in producing AuNPs of different shapes and sizes (6).

5.3.1.3 Green Synthesis

Green synthesis is an environmentally friendly method as it uses biological entities, that is, plant extracts, bacteria, or fungi, as reducing and stabilizing agents, giving biocompatible nanoparticles. The characteristics of the AuNPs synthesized this way can vary considerably depending on the biological source. For example, in 2004, Shankar et al. synthesized stable and biocompatible AuNPs with an aqueous leaf extract of *Azadirachta indica* (neem) (32).

5.3.1.4 Physical Methods

Techniques such as laser ablation take advantage of high-energy laser pulses to form bulk gold AuNPs without chemical agents, leading to pure nanoparticles. By changing the laser parameters, the size and shape of the AuNPs can be controlled. It was found that the laser ablation for synthesizing AuNPs showed a highly clean synthetic for getting chemically uncontaminated nanoparticles (33).

Each different synthesis route will impact the size, shape, surface chemistry, and stability of the AuNPs and, therefore, their potential for biomedical applications. Chemical reduction typically results in spherical nanoparticles with good dimensional uniformity, while seed-mediated growth allows for the synthesis of various morphologies (such as rods, cubes, or plates). Green synthesis generally yields biocompatible nanoparticles of different shapes, whereas physical methods produce pure nanoparticles suitable for sensitive applications.

The unique optical properties are one of the most outstanding features of AuNPs in their biomedical applications because they derive from SPR. SPR occurs when the conduction electrons at the surface of AuNPs oscillate in response to light, facilitating strong light absorption and scattering (34).

5.3.1.5 Imaging and Diagnostics

Intense light scattering and absorption phenomena associated with imaging and diagnostics greatly aid AuNPs in imaging applications, for example, dark-field microscopy gives high contrast while imaging biological specimens. Additionally, AuNPs may also be functionalized for targeting specific biomarkers in the early diagnosis of the disease. Jain et al. imaged cancer cells using AuNPs to provide contrast in the images (35).

5.3.1.6 Photothermal Therapy

Because AuNPs can absorb light and convert it to heat, they can be used in this method, which kills explicitly cancerous cells without affecting the surrounding tissues by just focusing directly on the targeted cancer cells. In a study by Hirsch et al., it was demonstrated that, in principle, tumor photothermal therapy with AuNPs is feasible (36).

5.3.1.7 Biosensing

Owing to the sensitive response of SPR to environmental conditions, it was readily applied to the sensing of biological entities. The SPR peak shifts in terms of color or optical features upon binding biomolecules to the AuNPs. In 1997, Elghanian et al. generated a colorimetric DNA detection approach that used AuNPs, and in this study, for the first time, this method was shown to be sensitive and specific (37).

5.3.1.8 Drug Delivery

The optical properties of AuNPs can be used in controlled drug delivery; the drug molecules are released upon photoexcitation at specific wavelengths of light. Such methods offer accurate control of drug release sites and schedules. For example, Fomina et al., in 2012, focused on studying controlled drug release using light-triggered AuNPs (38).

The choice of synthesis method plays a crucial role in determining the properties and functionality of nanoparticles. Additionally, the SPR of AuNPs allows for their use in various biological applications, thus contributing to advancements in nanomedicine. This includes innovations in imaging, therapy, sensing, and drug delivery.

5.3.2 Functionalization and Biomedical Applications

Surface modification of AuNPs becomes essential in targeted therapy as it imparts specificity, stability, and biocompatibility. The process entails the direct modification of AuNPs with various ligands, functional groups, or biomolecules to create nanoparticle conjugates that can specifically target cells or tissues. This results in the effective release and delivery of therapeutic agents.

5.3.2.1 Thiol-Based Functionalization

A common way of functionalizing is through thiol ($-SH$) groups conjugated on the surface and binding to gold strongly in monolayer form. This allows the molecular attachment of molecules such as drugs, peptides, and antibodies. Thiolated PEG improves biocompatibility and prolongs circulation time by preventing protein adsorption and reducing clearance by the MPS. Gref et al. discovered that PEGylation exponentially improved nanoparticle pharmacokinetics (39).

5.3.2.2 Ligand Conjugation

A ligand refers to an entity that binds to a specific site of a material. Examples of ligands that can be conjugated on AuNPs to allow them to be delivered directly to the target site include antibodies, peptides, and small molecules. Targeting ligands that bind specifically to overexpressed receptors on target cells can accomplish this by either covalent or non-covalent attachment. One effective example is the folic acid conjugate to target cancer cells overexpressing folate receptors; such constructs will take up the same in higher amounts. For example, Mansoori et al. demonstrated the targeting of cancer cells with folic acid-conjugated AuNPs (40).

5.3.2.3 Aptamer Functionalization

Aptamers are short, single-stranded DNA or RNA molecules that bind highly to their target molecules through unique three-dimensional structures. The functionalization of AuNPs with aptamers provides a high degree of targeting. Aptamer-functionalized AuNPs can be used for molecular targeting of a range of biomolecules, including proteins and cell surface receptors. Song et al. showed improved targeting and the therapeutic impact of these aptamer-functionalized AuNPs in cancer treatment (41).

5.3.2.4 Peptide Functionalization

Peptides can be used to modify the surface of AuNPs for targeting and therapeutic applications. Cell-penetrating peptides, such as the trans-activator of transcription (TAT) peptide, have been effectively used to improve the cellular uptake of AuNPs, while targeting peptides, such as RGD (arginine–glycine–aspartic acid), can be used to target integrins overexpressed in cancer cells. Studies report that such peptide-conjugated AuNPs have been applied to imaging and targeted drug delivery. Qin et al. used RGD-modified AuNPs for targeted cancer therapy (42).

To summarize, the surface of AuNPs can be modified for targeted therapies using methods such as thiol-based functionalization, ligand conjugation, aptamer functionalization, and peptide functionalization. All these add to the specificity, stability, and therapeutic effectiveness of the AuNPs in targeted therapy. The high surface area, ease of functionalization, and biocompatibility make functionalized AuNPs potent drug and gene delivery platforms. Their main applications are summarized as follows:

5.3.2.5 Drug Delivery Targeting

The nanoparticle can be engineered to deliver drugs specifically into target cells, thereby reducing off-target effects while improving therapeutic efficacy. The surface of nanoparticles can be functionalized with a targeting ligand such as antibodies, peptides, or small molecules that specifically bind to specific receptors on the target cell. For example, AuNPs conjugated with antibodies showed efficient delivery of chemotherapeutic drugs specifically to cancer cells while being less toxic to normal tissues. AuNPs have also been found to be effective as carriers of doxorubicin, a DNA-interactive chemotherapy drug, against prostate cancer (43).

5.3.2.6 Gene Transfection

AuNPs have been extensively applied as carriers in supplying the genetic material for the transfection of DNA, RNA, or siRNA into cellular systems. The positive charge on the surfaces of the AuNPs can facilitate the formation of complexes with negatively charged nucleic acids. This interaction protects nucleic acids from degradation and enhances their uptake by cells. Additionally, AuNPs can be modified with CPPs or other substances to improve the transportation of genes. For example, AuNPs, when functionalized with the TAT peptide, can improve siRNA delivery into cancer cells (44).

5.3.2.7 Theranostics

Theranostic applications are perfectly applied to AuNPs because of therapy–diagnosis integration. That substantial optical property allows imaging techniques to be carried out; for example, in dark-field microscopy and surface-enhanced Raman spectroscopy (SERS), while their surface is capable of carrying therapeutic agents. This dual functionality can thus enable *in vivo* tracking of drug delivery and therapeutic effects. For instance, Kumar and Lim used AuNPs in 2023 to connect two strategies: photothermal therapy and imaging of cancer cells (45).

5.3.2.8 Controlled Drug Release

AuNPs can be synthesized to release drugs in the presence of specific stimuli, such as light or pH enzymes. One example is AuNPs that release drugs when exposed to near-infrared light; hence, the user can decide where and when the drugs should be released. This makes drug delivery more efficient and reduces side effects in an intelligent manner. Vines et al. (2019) reviewed the light-triggered anticancer drug delivery from AuNPs (46).

Essentially, the key uses of AuNPs in drug and gene delivery encompass targeted drug delivery, gene transfection, theranostics, and controlled drug release. AuNPs possess remarkable functionalization flexibility and distinctive optical properties, rendering them highly effective instruments in cutting-edge biomedical applications.

5.3.3 Challenges in Clinical Translation

The principle underlying the toxicity of AuNPs is based on such imperative criteria as size, surface chemistry, dosage, and accumulation in biological tissues. Despite their high potential and efficiency in various biomedical applications, all of the abovementioned factors may also promote negative biological responses, which in turn must be avoided.

5.3.3.1 Size and Shape

The size and shape of AuNPs significantly influence their biological interactions and toxicity. Smaller particles of AuNPs (<10 nm) can easily pass through biological membrane barriers and accumulate in tissues. This could result in toxicity in cells. Larger nanoparticles, however, are less capable of penetrating cells but may still produce injury based on bulk and surface interaction. Pan et al. showed that smaller AuNPs (2 nm) had a much higher level of cytotoxicity than the larger ones. In connection with this, size is a governing factor in establishing the toxicity of these metallic nanoparticles (47).

5.3.3.2 Surface Chemistry

AuNPs are highly engineered toward surface functionalization, and thus, they dramatically influence biocompatibility and interaction with biological systems. Functional groups and coatings, such as PEG, reduce toxicity by preventing protein adsorption and immune recognition. However, certain surface modifications may evoke adverse immune responses or cellular stress. For instance, Alkilany and Murphy (2010) found that uncoated AuNPs had a significantly higher cytotoxic effect than PEGylated ones in one *in vitro* study (48).

5.3.3.3 Dose and Duration of Exposure

Both the dose and the duration of exposure have a significant impact on the toxicity of AuNPs. They can induce oxidative stress, inflammation, and cell death with high concentrations and prolonged exposure. Some reports have shown that repeated or high-dose administration of AuNPs may lead to the accumulation of AuNPs in organs with potential toxicity. For example, intravenous treatment of AuNPs was found to cause significant accumulation in the liver and spleen, thereby leading to histopathological changes and toxicity in mice (49).

5.3.3.4 Bioaccumulation and Biodistribution

Generally, the main concerns with using AuNPs are their biodistribution and long-term fate within the body. Potential toxic effects of persistent AuNPs in organs include damage to organs such as the liver, spleen, kidneys, and lungs. In addition to the size effect, the clearance of AuNPs is also related to their surface properties, with smaller AuNPs generally being slowly cleared. Another biodistribution and clearance study on AuNPs by Khlebtsov and Dykman in 2011 pointed out the need for a comprehensive investigation of these aspects within preclinical studies (50).

In essence, major toxicity concerns about AuNPs include size and shape, surface chemistry, dosage, and potential bioaccumulation. Addressing these issues through responsible design and thorough preclinical testing is essential for the clinically safe translation of AuNP-based therapies.

In clinical translation, AuNPs suffer from many key regulatory challenges. The primary one is based on the fact that the area of nanomaterials is inherently complex, so safety and efficiency tests must be rigid. Significant and vital regulatory challenges are standardization, toxicity evaluation, long-term safety, and manufacturing consistency.

5.3.3.5 Standardization

Lack of standardization in protocols for characterization and evaluation is one of the crucial regulatory challenges for AuNP technologies. Indeed, the wide variation in size, shape, surface modification, and synthesis method detracts from such standardizations in AuNP technologies. Regulatory bodies like the Food and Drug Administration (FDA) and the European Medicines Agency (EMA) stress that there should be a comprehensive characterization of nanomaterials, including their physicochemical properties, stability, and purity. A study by Sharifi et al. (2022) elaborated on the need for standardized methods to provide reproducibility and reliability in the characterization of nanomaterials (51).

5.3.3.6 Toxicity Evaluation

A full toxicity assessment is an absolute must for regulatory approval of AuNPs. It should include both *in vitro* and *in vivo* studies assessing cytotoxicity, genotoxicity, immunotoxicity, and long-term effects. These biological interactions and the potential for unexpected toxic effects can be pretty complex. Regulatory agencies need a lot of data to prove that AuNPs do not pose undue risks to patients. As previously stated, a literature review by Fadeel et al. (2010) reported the need for the interdisciplinary development of robust toxicity testing frameworks for nanomaterials (52).

5.3.3.7 Long-Term Safety

Determining the long-term safety of AuNPs becomes somewhat complicated as these nanoparticles have the potential for bioaccumulation and prolonged retention within the body. Long-term studies are required to assess the chronic effects and delayed toxicity potential. Various regulatory agencies demand detailed pharmacokinetics, biodistribution, and excretion studies of AuNPs to understand the long-term behavior of this nanoparticle in biological systems. Indeed, the necessity for long-term safety studies was noted about the risks that can ensue from the persistent nature of the AuNPs by Khlebtsov and Dykman (2011) (50).

5.3.3.8 Manufacturing Consistency

The manufacturing consistency of AuNPs is critical for their successful clinical translation. Variations in synthesis methods and inconsistent batch-to-batch methods may affect the quality and performance of the synthesized AuNPs. Regulatory agencies call for strict measures about quality control processes that are put into action to ensure that developed nanoparticles meet pre-defined specifications. The formulation of protocols for good manufacturing practices for nanomaterials is essential to satisfy these regulatory standards. Nanoparticle scaling-up is not easy. According to Paliwal et al. (2014), it was not easy to scale up the production of nanomaterials without decreasing product quality and reproducibility (53).

5.3.3.9 *Regulatory Pathways*

The regulatory pathways for nanomedicines are complex; specific guidelines must be followed strictly about documentation. Regulatory agencies have developed some guidelines for developing and approving nanomedicines, but the dynamic nature of this field makes it a challenging task. Besides, all these challenges could be tackled when researchers, regulators, and industries work collaboratively to enable the clinical translation of AuNPs. On the other hand, issuing guidance for nanomaterials in drug products is one of the FDA's missions in applying these technologies to new drug product development (54).

In short, there are multiple challenges from a regulatory perspective in the clinical use of AuNPs: standardization, comprehensive toxicology evaluation, long-term safety assessment, manufacturing consistency, and complex regulatory pathways. Addressing such challenges with collaborative efforts and proper testing is key to the successful clinical translation of AuNP-based therapies.

5.4 IRON NANOPARTICLES

5.4.1 *Magnetic Properties and Synthesis*

The size, shape, crystallinity, and surface coating of FeNPs in synthesis have a significant impact on their magnetic properties. These characteristics play a crucial role in determining how these nanoparticles interact with magnetic fields and their suitability for various biomedical applications.

5.4.1.1 *Size and Shape*

The size and shape of FeNPs are critical to their magnetic properties. Nanoparticles with a size less than 20 nm typically exhibit superparamagnetic characteristics. Consequently, they lack remanence and coercivity, indicating that they do not maintain magnetization once the external magnetic field is removed. In this sense, one of the most critical aspects of developing FeNPs is the control of the synthesis conditions, which include temperature, time of reaction, and precursor concentration. For instance, Peng et al. (2006) demonstrated that, under appropriate control, reaction parameters in the thermal decomposition could lead to monodisperse FeNPs with improved magnetic properties (55).

5.4.1.2 *Crystallinity*

Another factor that determines the magnetic behavior of the FeNPs is crystallinity. Better crystallinity often results in better magnetic properties as there are fewer surface defects and the magnetic domains are more aligned. This can be improved during the synthesis process through, for example, the annealing temperature or time. For instance, Park et al. (2005) showed that nanoparticles synthesized at high temperatures have increased crystallinity, increased magnetic saturation, and decreased coercivity (56).

5.4.1.3 *Surface Coating*

The surface coating is a crucial step toward enhancing the stability and interaction of FeNPs with biological environments. The surface coatings of nanoparticles serve to prevent aggregation and maintain colloidal stability. These coatings can consist of silica, polymers, or surfactants. Additionally, they may possess functional groups that can be further modified. Foglia et al. (2017) demonstrated that silica-coated FeNPs retained superparamagnetic properties and, when dispersed in aqueous solutions, were biocompatible and stable (57).

5.4.1.4 *Synthesis Method*

The magnetic characteristics of the FeNP that is produced are influenced by the specific method used for synthesis, which can be coprecipitation, thermal decomposition, hydrothermal, or solvothermal synthesis. Coprecipitation can result in the formation of nanoparticles with varying sizes and magnetic properties (58). In contrast, thermal decomposition offers a more effective method for controlling the size and shape of particles, resulting in enhanced magnetic properties. Hydrothermal and solvothermal techniques can prepare high-quality, highly crystalline nanoparticles with superior magnetic properties. In a review by Laurent et al. (2009), different synthesis methods and their impact on the magnetic properties of FeNPs were discussed, as were their advantages and drawbacks with the techniques described (59).

The magnetic properties of FeNPs are significantly influenced by manipulating various factors, including size, shape, crystallinity, surface coating, and the choice of synthesis method. Optimizing these conditions is essential for achieving FeNPs with tailored magnetic properties, which are necessary for their application in specific biomedical contexts.

The distinctive magnetic properties of FeNPs play a role in various biomedicine applications of MRI, targeted drug delivery, hyperthermia therapy, biosensing, and MPI. These all depend on the ability of FeNPs to react to externally applied magnetic fields for better diagnostic and curative functions.

MRI is carried out by using SPIONs as contrast agents. Their high magnetic properties allow for an improved contrast of images from MRI by reducing the relaxation times of hydrogen protons located nearby, producing the best possible quality, highly detailed images of tissues and organs. Such an application greatly benefits detecting tumors, inflammation, and vascular diseases. The research by Weissleder et al. (1990) looked at how well SPIONs could improve the contrast of an MRI scan to find metastases in the lymph nodes (60).

5.4.1.5 Targeted Drug Delivery

The magnetic properties of FeNPs offer the development and exploitation of a targeted drug delivery system by applying an external magnetic field to direct FeNPs to desired body locations and concentrate therapeutic agents at the target site with minimal systemic side effects. In this manner, treatment efficacy is immense, especially in cancer therapy. Sun et al. (2008) reported using magnetically guided FeNPs for targeted delivery of anticancer drugs with increased therapeutic effects in tumor models (18).

5.4.1.6 Hyperthermia Treatment

Magnetic hyperthermia uses FeNPs for generating localized heating under exposure to an alternating magnetic field. The heat generated is then able to kill cancer cells or improve the effects of chemotherapy and radiotherapy. This treatment will be helpful because it will act on the specific zones without affecting normal tissues around them. Jordan et al. (1999) have presented how this magnetic hyperthermia using FeNPs can have a high potential for effective cancer therapy using a reliable technique (61).

5.4.1.7 Biosensing

FeNPs are also used for biosensing. To make them work, they are attached to specific ligands or antibodies that allow them to detect and capture biomolecules like proteins, nucleic acids, or pathogens. Additionally, FeNP magnetic responses enable magnetism-based detection systems to quickly and sensitively detect such targets. A research report by Natarajan et al. (2024) focusing on the functionalization of FeNPs for sensitive biomolecule detection has confirmed the high potential of such NPs in diagnostic techniques (62).

5.4.1.8 Magnetic Particle Imaging

MPI is a new imaging modality based on direct imaging of magnetization, which FeNPs carry. In contrast to MRI, MPI is completely radiation-free and does not depend on the relaxation time of protons, making it much more secure for these reasons. Due to its exceptional sensitivity and specificity, MPI can effectively capture real-time images of the *in vivo* distribution of FeNPs. The technique was initially proposed by Gleich and Weizenecker (2005), and its significant value for medical applications was subsequently demonstrated (63).

The various applications highlight the significance of the magnetic properties of FeNPs in improving diagnostic and therapeutic capabilities, thereby significantly contributing to the advancement of nanomedicine.

5.4.2 Functionalization and Applications

Functionalizing FeNPs for biocompatibility is a crucial strategy for their safe and efficient use in various biomedical applications. This involves strategies to enhance their compatibility with biological systems.

Coating the surface is one of the vital approaches to reducing FeNPs' toxicity and enhancing stability. Commonly used materials are PEG, with PEGylation involving the attachment of PEG chains to the surface of the nanoparticle to give it solubility and reduce immunogenicity, thus prohibiting elimination by the reticuloendothelial system (64). Dendrimers, on their part, are branched polymers used for functionalization with numerous ligands that have better biocompatibility and targeting efficiency, providing controlled surface functionality with precision (65). An alternative way is the silica shell, which makes them chemically inert and biocompatible; this can also be further functionalized with biomolecules for targeted delivery (66).

The original surface ligands on FeNPs are exchanged for biocompatible ones to have higher stability in biological environments and lower potential toxicity. Due to their property of forming a stable bilayer similar to the cell membrane, phospholipids decrease immunogenicity (67). The biopolymers chitosan and alginate coat FeNPs to make the interface biocompatible and biodegradable, thereby making the nanoparticles more stable in the surrounding environment (68).

Encapsulation surrounds FeNPs with a biocompatible matrix, enhancing biocompatibility and enabling controlled therapeutic agent release. Liposomes are vesicles capable of entrapping FeNPs to offer a stable and biocompatible environment. They can further be functionalized with targeting ligands to achieve site-specific drug delivery (69). Polymeric micelles, formed from amphiphilic block copolymers, encapsulate the FeNPs and improve solubility and stability in biological fluids (70).

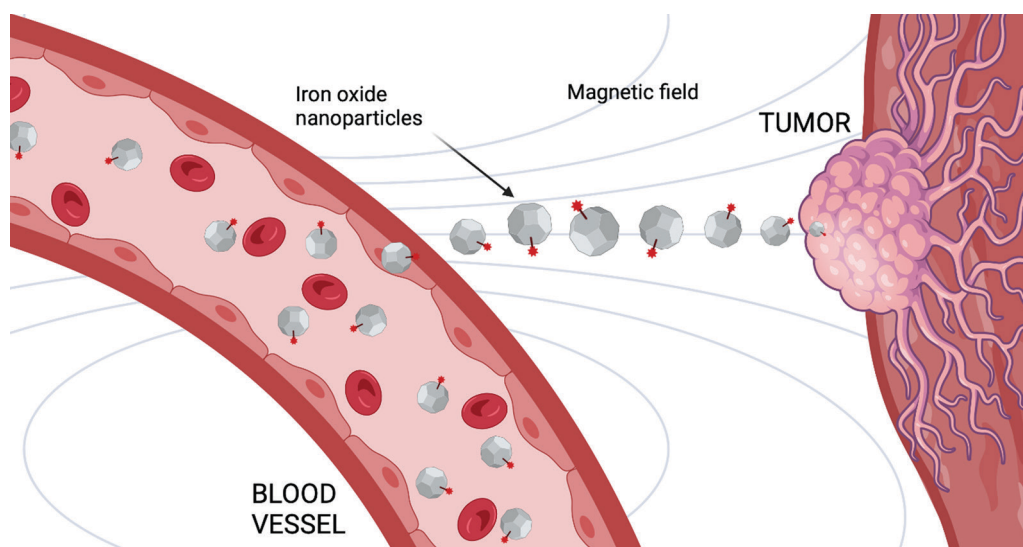


Figure 5.2 Magnetic targeting of iron oxide nanoparticles for tumor treatment. Innovative use of iron oxide nanoparticles in targeted cancer therapy. An external magnetic field guides these nanoparticles through a blood vessel, directing them towards a tumor site. As the nanoparticles flow alongside red blood cells, the magnetic force concentrates them on the tumor's location. This approach demonstrates the potential of magnetically guided drug delivery to enhance treatment efficacy by increasing local drug concentrations at the tumor while minimizing systemic exposure, thus potentially reducing side effects in cancer therapy.

These functionalization strategies are very important for the successful application of FeNPs in the biomedical arena, ensuring their safe and effective use in the clinical setting.

The characteristic magnetic properties of FeNPs find utility in targeted drug delivery and hyperthermia treatments. In a magnetic targeting system, an external magnetic field is applied to target FeNPs toward sites of interest in the body to increase the precision of drug delivery and lower systemic side effects. This can ensure high local drug concentrations at the target site, leading to huge improvements in treatment outcomes and consequent side effect minimization. For instance, targeted delivery of chemotherapeutics using FeNPs has been shown to yield promising results in reducing tumor sizes with minimal side effects (15). FeNPs can also enhance MRI contrast for better imaging of tissues to monitor targeted delivery processes (71) (Figure 5.2).

In hyperthermia applications, FeNPs generate localized heat under the action of an alternating magnetic field used in the selective ablation of cancer cells. This method can enhance the effectiveness of conventional therapies such as chemotherapy and radiotherapy. The heat generated by FeNPs increases the permeability of tumor cells to chemotherapeutic agents, improving drug uptake and treatment effectiveness (72). Clinical trials have demonstrated the potential of magnetic hyperthermia as an adjunct therapy for treating some types of cancer, highlighting its significant advantages in cancer treatment (73).

Generally, with these advanced functionalization and application techniques, it is evident that FeNPs are potential game changers regarding the treatment of cancer, making huge strides in nanomedicine.

5.4.3 Challenges and Opportunities

FeNPs have increasingly been recognized for their potential in various biomedical applications. However, challenges pertaining to their biodegradability and clearance from biological systems act as great obstacles to the safe use and efficiency of these FeNPs.

Most of the time, FeNPs are chemically stable, thus making them persist for a long time inside biological systems. The inert nature of some coatings applied to increase the performance of the coating can slow down the natural degradation processes and cause the possibility of environmental accumulation and ecological risks (7). The nanoparticles can accumulate in the water systems and in the soil, where they may impact different organisms, disrupting ecosystems (74). Moreover, the non-biodegradable FeNPs can accumulate in critical organs like the liver, spleen, and lungs, which may cause long-term toxicity. This accumulation may lead to oxidative stress, inflammation, and cellular damage (75).

The clearance of FeNPs also poses a significant problem. They are quickly taken up by reticuloendothelial system (RES), especially in the liver and spleen, creating a potential for toxicity in these organs (76). Moreover, their size often causes the FeNPs to avoid renal clearance, which has been efficient only for particles that are less than 5–6 nm, therefore increasing the circulation time in the blood and the potential for toxicity (77). Even desirable surface modifications, such as PEGylation for increased biocompatibility and longer circulation times, may impede the clearance processes, thus increasing their bioaccumulation (78).

In view of these limitations, several strategies have been designed, including developing biodegradable coatings, which break down into non-toxic products, toward enhanced clearance from the body. Materials such as dextran, albumin, and other biodegradable polymers are being explored to make this process easier (79). Similar considerations would include better engineering of size and shape for FeNPs to enhance renal clearance; for example, rod-shaped particles often exhibit higher clearance than sphere-shaped particles with the same volume and surface area characteristics (64). Surface functionalization with ligands that enhance clearances through specific cellular pathways would also be an improvement. For example, macrophage-targeted ligands may be more successful in enhancing cellular internalization of FeNPs, followed by lysosomal degradation via phagocytosis (80).

Magnetic targeting technology utilizes the magnetic properties of FeNPs to direct them to specific sites within the body to improve treatment specificity and delivery. Some disadvantages exist, however: The efficiency of magnetic targeting depends significantly on the strength and gradient of the external magnetic field. The use of high-field strengths to guide FeNPs into deep tissue targets can, however, cause technical difficulties and safety issues as these would exert sufficient force on the particles. Furthermore, FeNPs could aggregate in the presence of a magnetic field, leading to less efficient targeting and potentially causing blockages in the vascular system (81). On the other hand, for the same reason, there is an urgent need to ensure that FeNPs are retained specifically at the target site and cause minimal effects on surrounding tissues (82).

Recent technology in magnetic targeting will try to improve such a drawback: gradient magnetic fields have provided the opportunity for enhanced precision of the targeting with FeNPs via improvements in the spatial dispersion of the field (83). New nanoparticle coatings, which would respond predictably to magnetic fields without causing the particles to aggregate, would improve target accuracy. For instance, polymers that become thermosensitive and therefore change their properties with temperature changes caused by a magnetic field have been shown to hold much promise in a number of recent studies (84). In addition, with aggregation inhibition and improvements in targeting accuracy, the method can reduce off-target effects and obtain high specificity during its integration with other methods like ligand-based targeting. Such multimodal approaches combine the advantages of magnetic and biological targeting mechanisms to improve the specificity of targeting (85). The size and shape of the nanoparticles must be well-designed for effective responsiveness to magnetism and subsequent biodistribution. According to these studies, rod-shaped nanoparticles have been found to align much better with the field orientation compared to spherical nanoparticles and hence effectively allow for better targeting (86).

In summary, challenges of biodegradability and clearance are of importance for the safe and effective use of FeNPs in any biomedical application. Continued innovation in magnetic targeting technologies, including magnetic field design, strategies for functionalizing nanoparticles, and multimodal targeting, holds great promise for overcoming present limitations and increasing the clinical applicability of these advanced materials.

5.5 SILICA NANOPARTICLES

5.5.1 Synthesis and Characterization

SiNPs have evoked great interest in nanotechnology because they are versatile and have a broad application spectrum, mainly in drug delivery systems. The basic issues with using SiNPs are size and porosity; these really vary in terms of their performance in different applications. Therefore, a lot of care is taken in synthesizing these with exact control over structural parameters. Sol–gel is one of the techniques most frequently used in the synthesis of SiNPs. It involves the hydrolysis and condensation of silicon alkoxides, such as tetraethyl orthosilicate (TEOS). By controlling factors such as pH, temperature, and reactant concentration, researchers can finely modulate the size and porosity of the nanoparticles, as given by the seminal work by Stöber, Fink, and Bohn in 1968 (87). The other method is the microemulsion method, where SiNPs are synthesized in micelle confines; surfactant concentration also dictates the size of the micelles and, hence, the size of the nanoparticles produced. This method is known to produce SiNPs that are uniform and monodisperse (88). Organic or inorganic materials are used in template-assisted methods of shaping SiNPs. Most commonly, surfactants like CTAB are used for synthesizing the corresponding MSNs as templates. The template concentration and type may further be adjusted to fine-tune nanoparticle porosity and size (89).

The burning of silicon precursors in a flame during flame synthesis provides a scalable approach to SiNP production, where the flame temperature, precursor concentration, and residence time are just some of the many critical parameters for nanoparticle size control (90).

A few factors affecting the size and porosity of SiNPs are briefly outlined here. Higher concentrations of silicon precursors usually result in the formation of larger nanoparticles because of the increased nucleation and growth rates. The pH and temperature of the reaction mixture can also affect hydrolysis and condensation in the sol-gel process. Besides, the concentration of the surfactant used in the microemulsion and template-assisted methods is very important for the size and porosity of SiNPs. Modification of the surface of SiNPs must provide drug loading and release sites. Organic molecules with amine groups, thiols, or carboxyl groups increase the drug-loading capability of the nanoparticles. Aminated SiNPs create electrostatic interactions with negatively charged drug molecules, increasing loading efficiencies (19). Coating with biocompatible polymers, such as PEG, chitosan, or poly(lactic-co-glycolic acid) (PLGA), not only maximizes the potential for drug loading but can also adjust drug release rates. The hydrophilic environments created by the polymer coatings help stabilize hydrophobic drugs, which would otherwise undergo aggregation. By incorporating a slowly degrading matrix into the coat, drug release is prolonged (91). Inorganic coatings, for instance, calcium phosphate or gold, may also add drug-loading enhancements and serve as controlled release mechanisms through the gradual dissolution or disintegration of the coating in many forms (8). Increasing the number of available binding sites on the SiNP in surface modification causes an increase in the drug-loading capacity. These can also be designed to respond to specific environmental stimuli, such as pH and temperature, for controlled release. For instance, acidic cleavage of polymer-coated SiNPs allows their degradation within tumor tissue, providing targeted drug release (92). In addition, functionalization of SiNPs with biocompatible materials decreases toxicity and increases biocompatibility, hence improving their blood circulation time (93). By coating SiNPs with ligands like antibodies or peptides, it is possible to further improve targeted delivery and reduce unintended side effects (94). This necessitates careful SiNP synthesis and rational surface modification to be carried out to realize the potential of SiNPs in drug delivery. In this way, not only do these approaches refine the structural features of the nanoparticles, but they also improve their drug loading, release properties, and overall biocompatibility. The same confirms that SiNPs are a very versatile platform for advanced therapeutic applications.

5.5.2 Biomedical Applications

SiNPs have gained significant interest in nanotechnology for their potential in drug delivery systems due to their unique properties, such as their large surface area, tunable pore size, and ease of surface functionalization. These characteristics render SiNPs ideal for applications in sustained and controlled drug release. SiNPs feature a mesoporous structure that is conducive to the encapsulation of a wide variety of drugs within their pores. By adjusting the pore size, researchers can effectively load different drug molecules and control their release rate. Tailoring the pore diameter enables regulation of the drug's diffusion rate, providing sustained release over prolonged periods. This ability to modulate release kinetics is crucial in applications like chronic disease management, where maintaining consistent drug levels is essential (19).

Additionally, surface functionalization of SiNPs allows for further control over drug release. Coating SiNPs with PEG creates a hydrophilic barrier that moderates the release of hydrophobic drugs, yielding a controlled release profile. Moreover, employing pH-responsive or temperature-sensitive polymers can achieve stimulus-triggered drug release, activated by specific physiological conditions (8). Another approach involves the encapsulation of drugs within SiNPs or the creation of core-shell structures where the core contains the drug and the shell controls its release. This method provides an added layer of control, safeguarding the drug from premature release and ensuring its delivery at the intended site (95).

In cancer therapy, SiNPs are utilized to deliver chemotherapeutic agents in a controlled manner. Doxorubicin-loaded SiNPs, for example, provide sustained release of the drug, maintaining therapeutic levels over an extended time frame, thereby reducing dosing frequency and minimizing side effects. This approach enhances treatment efficacy and patient compliance (3). SiNPs are also employed in the controlled release of antibiotics. Encapsulating antibiotics within SiNPs achieves prolonged release profiles, helping maintain effective drug concentrations at infection sites and reducing the risk of resistance development (96).

SiNPs are highly valued in diagnostics and imaging due to their unique optical properties, high surface area, and capability to be functionalized with various biomolecules, making them efficient contrast agents and probes for different imaging techniques. In fluorescence imaging, SiNPs can be doped with fluorescent dyes or QDs, enhancing fluorescence signals, while the silica matrix protects the dyes from photobleaching. This property is particularly useful for stable and long-lasting imaging applications. For instance, SiNPs doped with rhodamine B have been effectively used for *in vivo* imaging of tumors, offering detailed visualization of cancerous tissues (97).

For MRI, FeNPs coated with silica shells serve as contrast agents, enhancing soft tissue visualization and detecting abnormalities with improved clarity. The silica coating's biocompatibility ensures safe *in vivo* usage (98). In computed tomography (CT) imaging, modifying SiNPs with elements such as gold or bismuth enhances X-ray attenuation, providing superior contrast compared to traditional agents and allowing detailed visualization of vascular structures and tumors (99).

SiNPs also excel in molecular diagnostics. They can be functionalized with antibodies, enzymes, or nucleic acids to create sensitive and specific biosensors for detecting various analytes, including proteins, DNA, and small molecules. These capabilities make SiNPs valuable for disease diagnosis and monitoring. For example, SiNP-based biosensors developed for cancer biomarker detection offer rapid and accurate diagnostics (100). Furthermore, the functionalization flexibility of SiNPs facilitates the development of multiplexed assays, enabling simultaneous detection of multiple biomarkers or pathogens in a single test. This high-throughput capability is invaluable in clinical diagnostics and research, thereby enhancing the efficiency of medical testing (101).

In summary, SiNPs play a pivotal role in controlled drug release, diagnostics, and imaging, offering enhanced drug delivery, contrast, and diagnostic accuracy across a range of medical applications. Their ability to be meticulously tailored and functionalized underscores their versatility and immense potential for advancing healthcare technologies.

5.5.3 Challenges and Regulatory Aspects

SiNPs are being increasingly used in biomedical applications due to their unique properties; however, their long-term biocompatibility and toxicity are critical concerns that need to be thoroughly addressed to ensure their safe application.

The potential for chronic toxicity arises when SiNPs accumulate in organs like the liver, spleen, and lungs, leading to adverse effects such as inflammation, fibrosis, and oxidative stress, potentially causing organ damage and dysfunction. This accumulation highlights the need for careful consideration of SiNPs' long-term presence in the body (102). SiNPs can elicit an immune response, with the potential to cause immunotoxicity. The immune system may recognize SiNPs as foreign particles, initiating an inflammatory response that could lead to tissue damage and possibly the onset of autoimmune diseases (103). Studies have raised concerns about SiNPs' ability to cause DNA damage, which could lead to mutations and potentially contribute to carcinogenesis. The potential for SiNPs to induce oxidative DNA damage and interfere with the cell cycle underscores the importance of genotoxicity assessments in their evaluation (104).

To improve SiNPs' safety profile, surface modification techniques are employed. Coating SiNPs with biocompatible polymers such as PEG can minimize protein adsorption and immune recognition, thereby enhancing circulation time and reducing the risk of an adverse immune response (105). Developing biodegradable SiNPs that can be safely metabolized and excreted from the body could mitigate long-term toxicity risks. Researchers are exploring biodegradable materials for SiNPs that degrade into non-toxic byproducts to enhance safety (96). Thorough preclinical testing, including long-term *in vivo* studies, is crucial for assessing the safety and biocompatibility of SiNPs. Such studies should comprehensively evaluate the potential for chronic toxicity, immunotoxicity, and genotoxicity to ensure SiNPs' suitability for human use (106).

Standardizing the manufacturing processes of SiNPs is crucial for their reproducibility, scalability, and regulatory approval. Batch-to-batch variability, scalability challenges, and establishing robust quality control measures are major hurdles in the standardized production of SiNPs. Batch variability often results from differences in synthesis parameters such as reactant concentrations, temperature, and reaction time, which can alter SiNPs' properties like size, shape, and surface characteristics. Implementing Good Manufacturing Practices (GMP) can help minimize such variability by ensuring consistent, controlled production processes (107). Scaling up SiNPs' production while maintaining quality is difficult, as laboratory synthesis methods may not translate directly to industrial scales. Techniques such as flame synthesis and continuous flow reactors are promising for large-scale production but require further optimization to ensure consistent nanoparticle quality across different production scales (108).

Standardized characterization techniques are essential for the consistent quality assessment of SiNPs. Techniques like dynamic light scattering (DLS), transmission electron microscopy (TEM), and Brunauer–Emmett–Teller (BET) analysis should be standardized to provide reliable data on nanoparticle properties (109). Furthermore, establishing clear regulatory frameworks is critical for ensuring the safe production and application of SiNPs. Regulatory guidelines are evolving, with agencies like the FDA and EMA developing robust standards that address the unique challenges posed by nanomaterials, including SiNPs (110).

In summary, addressing the challenges of long-term biocompatibility, toxicity, and standardization in the manufacturing of SiNPs is crucial for leveraging their full potential in biomedical applications. Through

strategic modifications, comprehensive testing, and adherence to regulatory standards, SiNPs can be safely integrated into future medical technologies.

5.6 QUANTUM DOTS

5.6.1 Optical Properties and Synthesis

QDs are nanocrystals made of semiconductors that possess distinctive optical characteristics, specifically photoluminescence. These attributes make QDs extremely beneficial for biological imaging purposes. The photoluminescent characteristics of QDs arise from quantum confinement processes, wherein the size of the material determines its electrical properties. QDs, when stimulated by a light source, produce light at distinct wavelengths depending on their dimensions, composition, and surface properties. The ability of QDs to adjust their emission wavelength allows them to produce a wide range of colors, spanning from ultraviolet to near-infrared. This versatility makes QDs valuable for many imaging applications in biomedicine (111, 112).

The exceptional brightness and photostability of QDs enhance the field of biomedical imaging. Contrary to conventional organic dyes, QDs have a lower tendency to lose their color when exposed to light, which enables longer imaging sessions and repeated exposure without a noticeable decrease in signal strength. This characteristic is especially advantageous in the context of live-cell imaging and long-term tracking research. Moreover, the capacity to combine QDs with biomolecules like antibodies and peptides enables precise visualization of particular cellular structures and processes (113, 114).

The production of QDs entails techniques that meticulously regulate their dimensions, morphology, and surface composition to attain the intended optical characteristics. The most prevalent technique for producing QDs is colloidal synthesis, which involves breaking down precursor materials in the presence of surfactants that restrict particle growth. This process provides monodisperse QDs with high quantum yields. Additional techniques encompass molecular beam epitaxy and chemical vapor deposition, which provide meticulous manipulation of QD composition and structure, albeit at a higher level of intricacy and expense (115).

The size of QDs is inherently connected to their optical properties as a result of the quantum confinement effect (116). As the QD size decreases, the difference in energy between the valence and conduction bands grows, causing the emission wavelength to shift toward the blue end of the spectrum (117). The ability to adjust the properties of QDs enables the creation of QDs with precise emission characteristics that are well-suited for a range of applications, including bioimaging and photovoltaic devices (118).

Size-dependent features play a vital role in enhancing the performance of QDs in various environments. Smaller QDs generally exhibit greater surface-to-volume ratios, which in turn affect their chemical reactivity and stability (119). Surface modifications, such as ligand attachment, are frequently used to enhance solubility, biocompatibility, and functionality (120). These alterations not only provide stability to the QDs but also enable precise adjustment of their optical characteristics to fulfill certain demands.

Several synthesis techniques have been devised to create QDs with accurate control over their size. Hot-injection techniques, such as the fast injection of precursors into a heated surfactant solution, facilitate the controlled formation and development of QDs (121). Solvothermal and hydrothermal procedures provide alternative techniques, frequently yielding high-quality QDs with consistent size distributions (122). In addition, continuous-flow microreactors have become a potential method for producing QDs on a large scale. They offer improved control over reaction conditions and enhance repeatability (123).

5.6.2 Biomedical Applications and Functionalization

In the biomedical field, QDs have emerged as a versatile instrument, particularly for the delivery of targeted medications (124). Bioconjugation with biological molecules, including nucleic acids, antibodies, and peptides, frequently leads to the functionalization of QDs, thereby increasing their utility (125). The covalent attachment of biomolecules is facilitated by the modifiable surface chemistry of QDs, which can be amended to include functional groups such as carboxyl, amine, and thiol groups (126).

Peptide-functionalized QDs are an exceptional choice for precision drug delivery due to their exceptional specificity in targeting cell receptors. For instance, RGD peptides can be conjugated with QDs to target integrin receptors that are overexpressed in tumor cells. This conjugation ensures that malignant tissues are precisely targeted, thereby significantly reducing off-target effects. In the same vein, antibodies, which are renowned for their exceptional affinity and specificity for antigens, are an additional popular option for QD functionalization. Drug delivery and imaging can be facilitated by antibody-conjugated QDs, which can precisely target specific proteins or cells (127, 128).

Additionally, QDs can be attached to nucleic acids, including DNA and RNA, for the purpose of biosensing and gene therapy. DNA-functionalized QDs have been implemented to monitor and visualize genetic material within cells, thereby providing valuable insights into molecular cell processes. Additionally,

these bioconjugated QDs have the capacity to facilitate the delivery of therapeutic nucleic acids to specific cells, thereby offering potential remedies for genetic disorders (129).

The biomedical imaging, sensing, and therapy fields have been transformed by the superior optical properties of QDs, which include high luminosity, photostability, and size-tunable emission spectra. Both *in vitro* and *in vivo* imaging applications are implemented with QDs. Due to their brilliant and stable fluorescence, they are the most suitable for the labeling and tracing of biomolecules, tissues, and cells over extended periods. QDs have been effectively implemented to visualize and monitor the behavior of cancer cells in real time (130).

In the discipline of biosensing, fluorescent instruments, such as QDs, are capable of detecting a wide range of biological molecules with high sensitivity and specificity. When conjugated with molecules that recognize specific analytes, QDs are capable of detecting biomarkers associated with a variety of diseases. This is crucial for the early diagnosis of the disease (131).

Drug delivery and photodynamic therapy are currently being investigated as therapeutic applications for QDs. Their ability to be functionalized with therapeutic agents and target-specific molecules facilitates the precise delivery of pharmaceuticals to diseased cells, thereby reducing systemic toxicity (132). Upon light activation in photodynamic therapy, QDs have the capacity to directly eradicate cancer cells by generating reactive oxygen species (133). This method offers a treatment option that is less invasive and has fewer adverse effects than conventional therapies.

In addition, the multifunctionality of QDs enables theranostics, a therapeutic and diagnostic approach that integrates their use (134). QDs are capable of administering therapeutic agents and diagnosing and monitoring the progression of diseases, thereby offering a comprehensive solution for disease management (135).

5.6.3 Challenges in Clinical Use

There are substantial obstacles to the clinical application of QDs as a result of environmental concerns and uncertainty regarding their toxicity (136). Numerous QDs are contaminated with heavy metals, including cadmium, lead, and selenium, which present toxicity hazards to living organisms. These metals have the potential to leach out of the QDs under physiological conditions, which could result in cytotoxicity and genotoxicity. For example, the release of cadmium ions from CdSe QDs has been demonstrated to result in oxidative stress, inflammation, and cell mortality, which pose significant risks for *in vivo* applications (137). In addition to their direct cytotoxicity, there are significant apprehensions regarding the long-term environmental consequences of QDs. The potential for bioaccumulation and the persistence of these substances in the environment raises concerns regarding their ecological toxicity. QDs have the potential to infiltrate the food chain and affect a variety of organisms by contaminating water sources and soil once they are released (138). Consequently, there is an urgent requirement to create safer QDs that are either more biodegradable or reduced in toxicity.

To reduce the toxicity of QDs, researchers have investigated a variety of strategies, such as the use of core-shell structures. In these structures, a less toxic shell material, such as zinc sulfide (ZnS), encases the toxic core, thereby reducing the discharge of harmful ions. Surface modifications with biocompatible polymers and ligands can also improve the biocompatibility of QDs. For instance, the interaction between biological tissues and QDs is diminished by encapsulating them with PEG, thereby reducing their toxicity.

The clinical efficacy of QDs is significantly influenced by their stability and aggregation in biological environments. QDs are susceptible to aggregation as a result of interactions with proteins and other biomolecules under physiological conditions, which can substantially impact their optical properties and biodistribution (12). Aggregation can result in the rapid clearance of QDs from the body or their unintended accumulation in non-target tissues, thereby compromising their efficacy and safety, as well as a decrease in their fluorescence intensity and stability (4).

There are numerous methods for improving the stability and averting the aggregation of QDs in biological environments. Surface functionalization with hydrophilic and biocompatible molecules, such as PEG, is an effective strategy that creates a steric barrier to prevent aggregation. The colloidal stability of QDs is preserved in physiological conditions through PEGylation, which guarantees consistent performance in therapeutic and imaging applications (139). Another effective approach to safeguarding QDs from the biological milieu is to encapsulate them in biocompatible matrices, or liposomes. These encapsulation techniques establish a physical barrier that protects QDs from aggregation and degradation while at the same time maintaining their optical properties (9). Furthermore, the durability of QDs in biological fluids can be improved by utilizing amphiphilic polymers to establish micelle-like structures around them (140). The integration of QDs with biological systems is enhanced using biomimetic coatings that mimic the cell membrane, which is a promising approach (141). These coatings not only enhance the biocompatibility

and targeting efficacy of QDs but also improve their stability, rendering them more suitable for clinical applications.

5.7 COMPARATIVE ANALYSIS OF INORGANIC NANOCARRIERS

5.7.1 Comparison of Physicochemical Properties

Inorganic nanocarriers' efficacy in the delivery of nucleic acids and proteins: a comparative analysis of their physicochemical properties. Inorganic nanoparticles, such as AuNPs, IONPs, SiNPs, and QDs, possess unique physicochemical properties that significantly influence their effectiveness in the delivery of nucleic acids and proteins. AuNPs are biocompatible, readily functional, and display unique plasmonic features through which effective conjugation with biomolecules like DNA, RNA, and proteins can be done. Their size and shape can be strictly controlled during synthesis to achieve improved cellular internalization and stability (142). IONPs are superparamagnetic and, hence, are used for MRI studies and magnetic targeting in drug delivery. Their surface could be derivatized with polymers or ligands to increase biocompatibility and reduce aggregation. The high potency of IONPs in gene delivery arises from their ability to form complexes with nucleic acids and to assist endosomal escape (143). SiNPs have a high surface area for easy functionalization and a porous structure. They offer a versatile platform for the loading and release of biomolecules for both hydrophilic and hydrophobic substance encapsulation. SiNPs can be engineered to degrade under certain conditions, making it possible to release therapeutic agents in a controlled manner (144). QDs possess unique optical properties like size-tunable emission and high photostability; hence, they have a wide application window in bioimaging and biosensing (145). However, a major drawback that remains with nanomaterials is their potential toxicity due to the release of heavy metals. Surface modifications, such as coating with biocompatible polymers, would be imperative for overcoming these drawbacks and rendering them safe for use in biomedical applications. The differences among these physicochemical properties are one of the key guiding factors toward the suitability of each nanoparticle type for specific biomedical applications. AuNPs are preferable for use because they are biocompatible, easy to functionalize, and have the best suitability for gene and protein delivery. IONPs are useful in applications such as magnetic targeting, imaging, and the delivery of therapeutics. SiNPs have largely passed that stage and are now used with excellent controlled release properties. QDs are mainly used in imaging and sensing, because of their optical properties.

Controlled release and stability are critical parameters for inorganic nanoparticles to be effectively applied in a biomedical setting. In comparison to other types of inorganic nanocarriers, the SiNPs show good stability and controlled release profiles. Because of their porous structure, hydrogels are able to easily encapsulate a variety of therapeutic agents, and surface modifications and environmental triggers like pH and temperature changes can control how quickly these agents release (146). AuNPs are known to be very stable because they are practically inert and do not oxidize easily. Functionalization with thiol-containing ligands can further increase the stability of AuNPs in biological environments. Responsive linkers can be designed to release their therapeutic agents at a site of interest based on specific triggers, such as changes in redox potential or the presence of enzymes (147). IONPs are stable under physiological conditions and can be functionalized for improvement of biocompatibility, such as decreasing aggregation; however, their release characteristics are less controlled compared to SiNPs and AuNPs. On the other hand, together with surface modifications, polymer coatings are also exploited to achieve not only stability but also the controlled release of conjugated biomolecules (148). QDs have problems related to their stability and potential toxicity. QDs can be modified using a wide range of biocompatible materials, such as PEG, to offer improved stability and reduce aggregation. QDs do not serve practical purposes for the controlled release of therapeutic agents; however, their most important application in imaging and sensing benefits from the stability provided by such surface modifications (149). In general, silica NPs embody the most synergistic combination of stability and controlled release features—making them the best candidate for most biomedical applications. Added to this is the huge degree of stability and freedom in their functionalization with AuNPs, whereas IONPs find a preference in applications with special magnetic properties. Following this are QDs, which, despite possibly having some toxicity, are still useful for imaging and sensing due to their distinctive optical properties (Figure 5.3).

5.7.2 Biocompatibility and Safety Profiles

The safety and biocompatibility of inorganic nanocarriers, like AuNPs, IONPs, SiNPs, and QDs, depend significantly on the type of composition, size, surface chemistry, and interaction with biological systems. The high biocompatibility of AuNPs derives primarily from their inert nature and low toxicity.

They are immensely stable and can be easily functionalized with biocompatible coatings, like PEG, to further increase their biocompatibility and reduce immune responses (150). IONPs, on the other hand, have moderate biocompatibility, mainly due to the surface modifications. Bare IONPs cause oxidative stress and

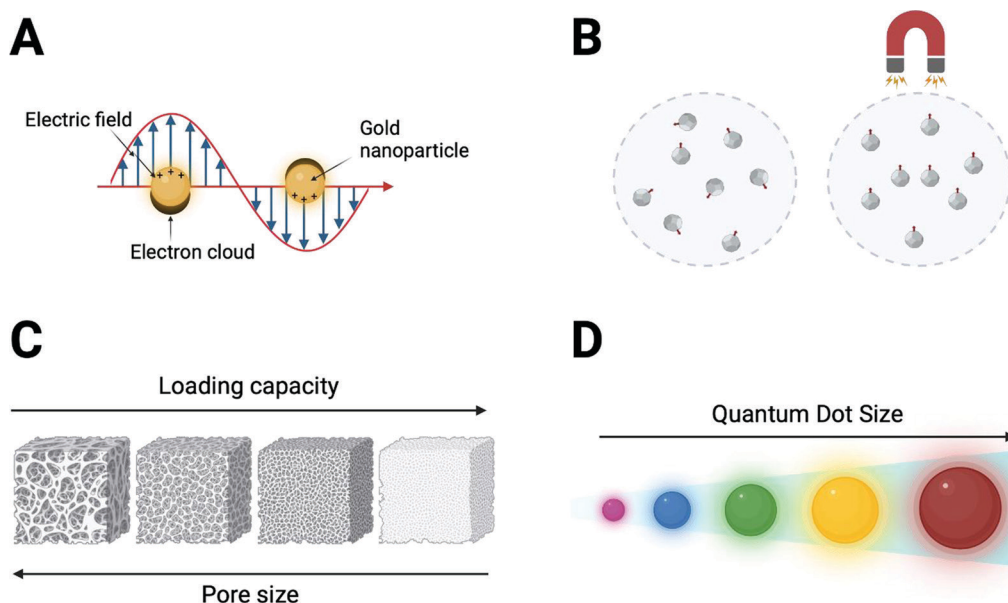


Figure 5.3 Key properties of inorganic nanocarriers. (A) Gold nanoparticles (AuNPs): They exhibit the surface plasmon resonance effect, showing the oscillation of the electron cloud in response to an electric field, which is key to their optical properties. (B) Iron oxide nanoparticles (IONPs): They respond to an external magnetic field, with nanoparticles aligning along the field lines, which is crucial for magnetic targeting and imaging applications. (C) Silica nanoparticles (SiNPs): They have a relationship between pore size and loading capacity. As pore size decreases from left to right, the available space for cargo loading increases, highlighting their potential for controlled drug delivery. (D) Quantum dots (QDs): They demonstrate size-dependent optical properties, where smaller quantum dots emit higher-energy (bluer) light and larger ones emit lower-energy (redder) light, essential for their use in imaging and sensing applications.

inflammation, but after coating them with some biocompatible polymers—like dextran or PEG—this safety profile improves significantly. Coating prevents aggregation and reduces interactions with proteins and cells, all of which make IONPs suitable for MRI and drug delivery purposes (151). SiNPs are biocompatible, especially if their surfaces are modified with agents that confer improved biocompatibility. Porous structures provide effective drug loading and controlled release, while surface functionalization with amino or carboxyl groups enhances its interaction with biological systems. However, there are doubts about some of the potential adverse effects of SiNPs, particularly in terms of long-term exposure, such as pro-inflammatory and pro-oxidative effects (152). QDs have shown distinct optical properties while, at the same time, causing significant biocompatibility challenges due to potential toxicity. Often containing heavy metals like cadmium and selenium, these metals can leach out of QDs and damage cellular machinery. Surface coatings with biocompatible materials like PEG or silica would reduce these, but long-term safety and the possibility of bioaccumulation remain a concern (153).

Each type of inorganic nanocarrier has specific toxicity concerns that must be addressed to ensure safe biomedical applications. Among the primary concerns with gold nanocarriers are immune response and cytotoxicity at higher concentrations. Mitigation strategies involve biocompatible polymers, such as PEG, which reduce protein adsorption and recognition by the immune system (154). On the other hand, IONPs cause oxidative stress and inflammatory responses because of their iron content, which catalyzes the formation of ROS. Coating IONPs with biocompatible polymers such as dextran, PEG, or silica tends to moderate this effect through a barrier that avoids direct interaction with biological tissues and reduces the release of iron ions (155). More generally, SiO_2 nanoparticles are considered to be biocompatible, but under high doses and prolonged exposure, they can trigger inflammation and oxidative stress. Surface functionalization with amino or carboxyl groups, or the use of mesoporous structures, decreases these adverse effects. Finally, the biodegradability of silica materials may be varied to minimize long-term toxicity (156). Highly toxic QDs containing heavy metals are considered serious as the leaching of cadmium or selenium ions from QDs causes cellular damage and severe toxic effects. Surface modifications with thick silica shells

or crosslinked polymers produce a robust coating, which can reduce the release of toxic ions and provide better biocompatibility. However, long-term studies are needed to fully understand QDs' environmental and health impacts (157).

In summary, although individual inorganic nanoparticles present unique advantages and challenges related to biocompatibility and safety, adequate surface modification associated with attention to their physicochemical properties can ease toxicity concerns and support their safe and effective use in biomedical applications.

5.7.3 Technological and Economic Considerations

One of the many important technological obstacles to increasing the production of inorganic nanocarriers, such as AuNPs, IONPs, SiNPs, and QDs, is ensuring that the size, shape, and surface properties of the particles remain consistent during large-scale syntheses. Nucleation and growth processes to synthesize AuNPs need to be very precisely controlled for uniformity, which dramatically becomes difficult in the industry on a large scale (158). For IONPs, ensuring high magnetic purity and stability while preventing oxidation and aggregation during the scale-up process is complex. Usually, the synthesis process goes through two steps: coprecipitation and thermal decomposition. Good control over these steps is required for the reproducibility of the process. In addition, surface coating with improved biocompatibility and functionalization increases the complexity of applying the coatings uniformly over large batches (159). Given the challenges posed by the pore size and functionalization of porous SiNP, it is crucial to maintain precise control over parameters such as pH, temperature, and reactant concentration during the synthesis process. This is necessary to ensure the reproducibility of both pore size and surface chemistry. These concerns, which are also mentioned in the literature, pertain to maintaining consistent results between batches and preventing clumping during storage and handling (160). Because QDs are composite, containing potentially toxic elements, careful control of the synthesis conditions is necessary to prevent the release of heavy metals. High-temperature processes, like organometallic synthesis, have to be carefully scaled to maintain the optical properties and stability of the QDs. Such surface coatings are also a significant challenge in providing safety and efficacy, free from toxicity and functional interference (161). These general challenges in scaling up nanoparticle production processes were extensively reviewed in 2023 by Fernandes et al. (162).

Economic considerations play a crucial role in the selection of inorganic nanocarriers for commercialization, with a specific focus on drug delivery. The cost of raw materials, synthesis, functionalization, and scalability are key to these nanocarriers' commercial feasibility. This is because AuNPs, although highly biocompatible and easily functionalized, are very expensive due to the expensive nature of the element itself. These result in the material's limited applicability, except when the benefits somehow surpass the cost in most commercial applications. The IONPs are relatively cheaper in terms of raw materials as iron is abundant and way cheaper than gold. On the other hand, high costs accrue from the synthesis process and the high level of surface modification needed for better biocompatibility and functionality. It is cardinal that the commercial values of the nanoparticles be balanced with cost, performance, and safety (163). On the other hand, SiNPs are relatively low-cost materials that offer great flexibility in both functionalization and controlled release properties. SiNPs are economically attractive for drug delivery applications, as compared to other materials, because silica is cheap. SiNPs have the benefit of simplicity and cost-effectiveness in the preparation of materials through the sol-gel synthesis methods. However, the cost of functionalization for ensuring batch consistency may just add to these expenses (164). Although there are unique optical properties of the QDs, they are, in most cases, prohibitively expensive due to their synthesis at high temperatures using complex procedures. They also need extended purification and surface modifications to prevent toxicity. Besides, the high production cost and stringent safety testing of the synthesis make QDs uneconomic in drug delivery (165). Therefore, in making this choice of inorganic nanocarriers for commercial drug delivery applications, the technological challenges as well as economic factors should be critically studied. High-performance AuNPs are expensive; IONPs and SiNPs provide economical solutions with moderate performance. Technologically advanced QDs pose a major economic problem in their commercial use. In the end, it all depends on specific application requirements, such as cost, performance, and the balance between those and safety.

5.8 FUTURE PERSPECTIVES AND EMERGING TRENDS

5.8.1 Advances in Material Science and Engineering

Recent advances in material science have opened up several new frontiers that have the potential to bring about radical changes in the functionality of inorganic nanocarriers. One of the most significant developments in this regard is the ability to engineer multifunctional nanoparticles capable of delivering multiple functionalities at the same time. For example, AuNPs have been designed for use as diagnostic and

therapeutic agents in a concept known as theranostics. The theranostic AuNPs will offer both functional imaging and therapeutic capabilities on the same platform, which is otherwise bulky and less efficient with current technologies (166). Also very promising are hybrid nanoparticles that are based on several inorganic materials. These hybrid structures combine the unique properties of each component, resulting in a synergistic effect that enhances overall performance. As an example, the silica-coated IONPs use the magnetic properties of iron and the biocompatibility and functionalization flexibility of silica to make them highly effective for drug targeting and MRI applications (167).

A further important development is associated with the use of surface modification techniques in an attempt to improve nanoparticle stability, biocompatibility, and target specificity. Adding PEG to the surfaces of nanoparticles has been shown to lower opsonization and increase blood circulation time, which makes drug delivery systems work better (168). New synthesis methods are critical for developing more effective and safer inorganic nanocarriers. These are generally the deficiencies, i.e., the traditional synthesis approaches are not reproducible, burdensome to the environment, and not scalable. However, the recent advances are trying to serve them all at once. Therefore, currently, it is almost on the brink of moving from basic research to practical applications.

One technique is to use green chemistry rules for nanoparticle synthesis. Green synthetic routes are based on biological agents like plant extracts, bacteria, and fungi in the eco-friendly synthesis of nanoparticles. This not only reduces the amount of toxic chemicals but also, in most cases, leads to nanoparticles with enhanced biocompatibility and less toxicity (169). Furthermore, microfluidic synthesis methods have been developed to provide controlled regulation over the size, shape, and composition of the nanoparticle. Microfluidic systems offer the highest level of homogeneity and reproducibility, which are very important in clinical applications. These systems also allow rapid screening of different synthesis conditions, thus accelerating development toward optimized nanocarriers (170).

Incorporation of these novel synthetic approaches can dramatically improve the safety profile of inorganic nanoparticles. For example, control over the size and surface characteristics of QDs makes them less potentially cytotoxic and more amenable to *in vivo* imaging applications. In the same way, magnetic features and coatings of FeNPs can be carefully adapted to mitigate side effects and improve therapeutic efficacy (171). These new synthesis methods in material science bring promising advancements for functionality and safety improvement in inorganic nanocarriers. That is, paramount developments would be realized in the potential of nanotechnology for applications in biomedicine and industry, consequently heralding new opportunities for diagnosis and treatments in medicine and beyond.

5.8.2 Integration with Other Nanotechnologies

The synergistic combination of inorganic nanocarriers with other nanotechnologies is likely to become a new paradigm in drug delivery. One such synergy is the production of nanocomposites, which blend nanoparticles with several organic and inorganic materials, providing an exclusive platform for multifunctional systems. As an example, drug loading into AuNPs incorporated with polymeric materials may enhance controlled release and targeting, hence potentiating the overall therapeutic index (172).

In addition, the hybrid systems resulting from combining these inorganic nanoparticles with other nanostructures, such as liposomes or dendrimers, may yield much better results. For example, liposome-coated QDs may combine the benefits of liposomal encapsulation, such as biocompatibility and prolonged circulation time, with the unique optical properties of QDs, thereby enabling simultaneous delivery and imaging. Such systems can be designed to respond to specific triggers, for instance, pH or temperature, in a way that encourages drug release only at sites of interest (173).

Computational modeling and machine learning are part of the transformation to *in silico* design and optimization of inorganic nanocarriers. Such capability allows the prediction and adjustment of nanoparticle properties toward desired therapeutic effects with a higher degree of precision and efficiency (174). On the other hand, machine learning further personalizes nanomedicine. By incorporating patient-specific information, such as genetic profiles and disease markers, machine learning will predict the behavior of different nanoparticle formulations in every individual patient. This personalized strategy ensures that treatment is in line with the needs of the individual patient, maximizing effectiveness while minimizing side effects (175).

With this, the convergence of inorganic nanocarriers with other forms of nanotechnology and further use of computational modeling and machine learning stand to seriously impact advances in drug delivery. All these novel breakthroughs will likely bring more precision, efficiency, and personalized treatments than ever before, ushering in a new era for nanomedicine.

5.8.3 Impact on Healthcare and Regulatory Issues

Advancements in inorganic nanocarriers have the potential to revolutionize several areas, mainly in the diagnosis and targeted therapy of healthcare. AuNPs, for instance, are being explored for the enhancement of imaging techniques and direct drug delivery to cancerous cells. Due to their high surface area, they are able to attach several ligands, enabling multiple druggings and leading to preferred target delivery with minimum side effects. The next is that they possess optical properties, which advance imaging and photothermal treatment, which is a dual application useful both in diagnosis and treatment (176).

Yet another example is FeNPs, specifically when focusing on MRI with respect to cancer hyperthermia treatment, due to their magnetic properties that facilitate better MRI contrast, hence ensuring an early and correct diagnosis. In addition, FeNPs can give out localized heat once subjected to an alternate magnetic field. This selectively kills the cancer cells and spares the surrounding healthy tissue (177).

However, several regulatory challenges have to be overcome before inorganic nanocarriers can be widely implemented in clinical applications. Among these is the comprehensive safety profiling that has to be done. This unpredictability in interactions with biological systems and other potential effects related to the specific properties of nanoparticles requires intensive toxicological testing. Long-term investigations should be carried out to study biodistribution, metabolism, and excretion, as well as any possible chronic effects (178).

Equally important, from a regulatory point of view, is standardization in the manufacturing processes because consistency in both nanoparticle synthesis and functionalization is a prerequisite for the reproducibility and reliability of clinical applications. Regulatory agencies need to establish guidelines for nanoparticle characterization and quality control that encompass the definition of parameters, size, shape, surface charge, and purity of nanoparticles (179).

One of the challenges is the regulatory classification of nanoparticle-based products. Traditional regulatory frameworks are possibly not fully fitted for most nanomedicines that typically reveal hybrid characteristics of drugs, biologics, and medical devices. It could help make clear guidelines and definitions that could enable easier regulatory approval and also assure that such products meet the necessary safety and efficacy standards (180).

On the other hand, developments in inorganic nanocarriers may strongly change healthcare, delivering new opportunities for improved diagnostics, targeted therapy, and precision medicine. Nevertheless, the successful translation of these technologies into clinical practice will depend on addressing the regulatory challenges with regard to their safety, manufacturing, and classification.

5.9 CONCLUSION

Inorganic nanocarriers, including AuNPs, FeNPs, SiNPs, and QDs, have shown significant promise in advancing various biomedical applications due to their unique properties. Each type of nanoparticle brings distinct advantages and faces specific challenges in medical treatment delivery. AuNPs are highly valued for their exceptional optical properties, particularly SPR, making them ideal for imaging and diagnostics. Their ability to be functionalized with various ligands and biomolecules enhances biocompatibility and targeting specificity in drug delivery systems. Moreover, AuNPs are effective in photothermal therapy, where light-induced heat is used to destroy cancer cells.

IONPs, known for their superparamagnetic properties, have transformed MRI by improving contrast and diagnostic accuracy. These magnetic properties also enable targeted drug delivery and hyperthermia treatment, where localized heating selectively kills cancer cells. SiNPs offer high surface areas and porous structures that enhance drug delivery mechanisms. Their biocompatibility allows for loading with therapeutic agents and controlled release, increasing therapeutic efficacy in applications like cancer therapy.

QDs are recognized for their size-tunable fluorescence and high photostability, making them valuable in bioimaging and diagnostics. Their bright, long-lasting signals are crucial for extended imaging experiments. However, the toxicity of QDs, primarily due to their heavy metal content, remains a significant concern.

The analysis of AuNPs, FeNPs, SiNPs, and QDs highlights that each nanoparticle type excels in different biomedical applications:

- *AuNPs*: Particularly effective in imaging and therapy, they play a key role in theranostics, which combines diagnosis and treatment on a single platform.
- *FeNPs*: Most suitable for magnetic applications like MRI and magnetically guided targeted drug delivery.
- *SiNPs*: Ideal for drug delivery due to their high surface area, biocompatibility, and controlled release capabilities.
- *QDs*: Excellent for imaging applications owing to their superior brightness and photostability, although their potential toxicity requires careful handling.

These findings are crucial for the broader field of nanomedicine, demonstrating the versatility and specific advantages of different inorganic carriers. The ability to customize nanoparticles facilitates precise site-specific drug delivery, leading to more potent therapeutics with fewer side effects. Additionally, advancements in synthesis and functionalization techniques aim to address existing issues like toxicity and stability, expanding the clinical applicability of these nanomaterials.

Future research should prioritize optimizing the synthesis and functionalization of inorganic nanocarriers to maximize therapeutic efficacy while minimizing toxicity. Developing hybrid systems that combine various nanoparticles could harness their individual strengths to address complex medical conditions more effectively. Furthermore, advancements in computational modeling and machine learning could accelerate the design and optimization of nanoparticles, enhancing their efficacy and safety in clinical applications.

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